

Analyse data and information to describe patterns, trends and relationships and identify anomalies

Learning intention

- analyse data and information to describe patterns, trends and relationships and identify anomalies.

Teach and model

- describing measures of central tendency such as mean, mode and median and identifying outliers for quantitative data.
- analysing changes in battery energy output following recharging over many cycles and relating to available chemical potential energy.
- comparing temperature differences obtained by reacting different proportions of the same chemicals to determine if there is a relationship.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.